



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

November 2011

Assessment Task 1
Year 11

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question if full marks are to be awarded.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 65

- Attempt sections A – D.
- Start each **NEW** section in a separate answer booklet.
- Hand in your answers in 4 separate bundles:

Section A
Section B
Section C
Section D

Examiner: *J. Chen*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

START A NEW ANSWER BOOKLET

SECTION A [15 marks]

For these 10 questions there is one correct answer per question. Write down in your answer booklet the question number and letter of your answer.

Marks

1.

[1]

$$\int \frac{1}{(1-2x)^3} \cdot dx$$

(a) $-\frac{1}{4(1-2x)^2} + C$

(b) $\frac{1}{4(1-2x)^2} + C$

(c) $-\frac{1}{2(1-2x)^2} + C$

(d) $\frac{1}{2(1-2x)^2} + C$

2. Find the acute angle between the lines $x + 3y = 0$ and $x - 2y = 1$.

[1]

(a) 135°

(b) 45°

(c) 22.5°

(d) None of the above

3. Solve $\cos\left(x - \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$, for $0 \leq x \leq 2\pi$.

[1]

(a) $x = 0, \frac{\pi}{3}, 2\pi$

(b) $x = \frac{\pi}{6}, \frac{11\pi}{6}$

(c) $x = \frac{\pi}{3}, 2\pi$

(d) $x = \pi, \frac{4\pi}{3}$

4. P and Q are the points $(-4, 3)$ and $(2, 1)$ respectively. What are the coordinates of the point M which divides QP externally in the ratio 4: 5.

[1]

(a) $M(26, -7)$

(b) $M(31, -3)$

(c) $M\left(-\frac{2}{3}, \frac{17}{9}\right)$

(d) $M(-28, 11)$

5. What is the exact value of $\tan 15^\circ$? [1]

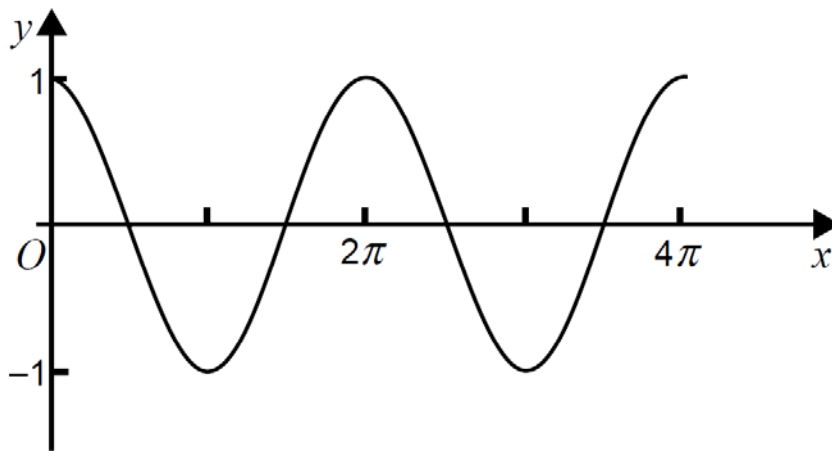
(a) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

(b) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

(c) $\frac{1-\sqrt{3}}{1+\sqrt{3}}$

(d) $\frac{1+\sqrt{3}}{1-\sqrt{3}}$

6. Part of the graph of a trigonometric function is shown below. [1]



A possible equation for the function is

(a) $y = \cos(2x)$

(b) $y = \cos x$

(c) $y = \sin(2x)$

(d) $y = \sin x$

7. The linear factors of $x^4 + x^3 - 3x^2 - 3x$ are [1]

(a) $x, x + \sqrt{3}, x - \sqrt{3}$

(b) $x + 1, x + \sqrt{3}, x - \sqrt{3}$

(c) $x, x + 1$

(d) $x, x + 1, x + \sqrt{3}, x - \sqrt{3}$

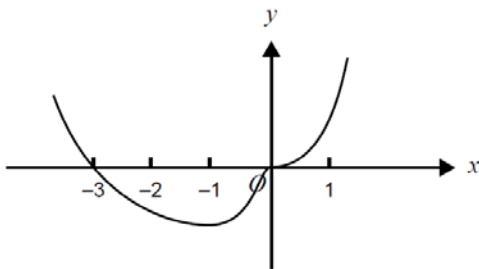
8. A continuous function $f(x)$ has the following properties.

[1]

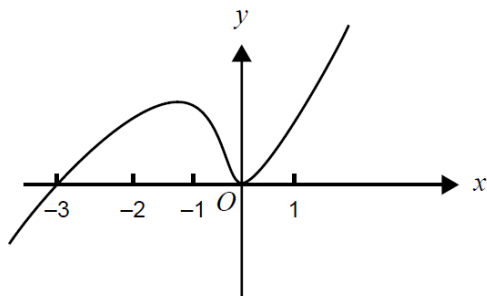
$$\begin{aligned} f(0) &= 0 \\ f(-3) &= 0 \\ f'(0) &= 0 \\ f'(-1) &= 0 \\ f'(x) &> 0 \text{ for } x < -1 \\ f'(x) &< 0 \text{ for } x > -1, x \neq 0 \end{aligned}$$

Which one of the following could represent the graph of $f(x)$?

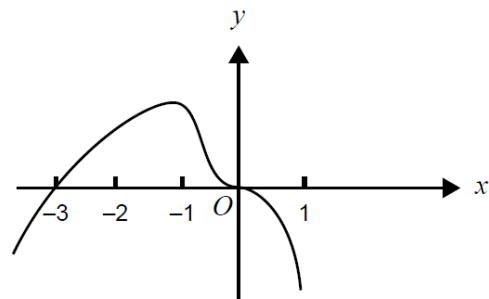
(a)



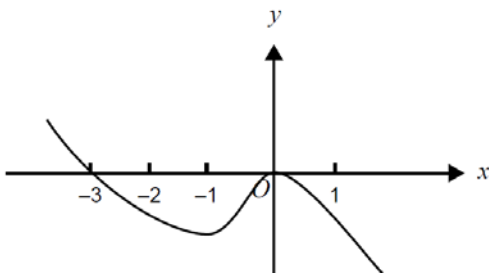
(b)



(c)

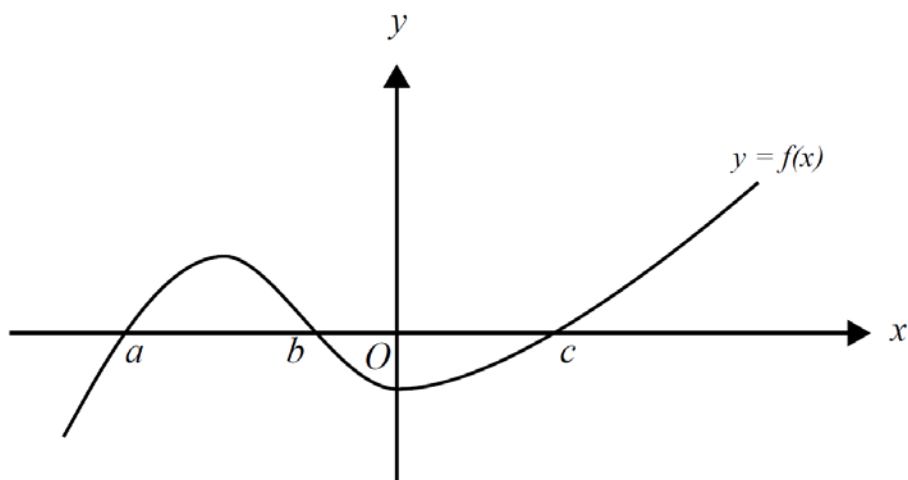


(d)



9. The diagram below shows part of the graph of a function $y = f(x)$.

[1]



The total area bounded by the function $y = f(x)$ and the x -axis between $x = a$ and $x = c$ is given by

(a)

$$\int_a^c f(x) \cdot dx$$

(b)

$$\int_a^b f(x) \cdot dx + \int_c^b f(x) \cdot dx$$

(c)

$$\int_a^b f(x) \cdot dx + \int_b^c f(x) \cdot dx$$

(d)

$$\int_b^c f(x) \cdot dx - \int_a^b f(x) \cdot dx$$

10. If $\int_0^4 f(x) \cdot dx = 3$, then $\int_0^4 (3f(x) - 2) \cdot dx$ is equal to

[1]

(a) 9

(b) 1

(c) 7

(d) 3

End of Multiple Choice Section

11. Find all angles θ for which $\tan \theta = \sqrt{3}$. [1]

12. Sketch the graph of a function $y = f(x)$ such that $\frac{dy}{dx} < 0$ and $\frac{d^2y}{dx^2} > 0$ for $0 \leq x \leq 2$. [2]

13. Prove that [2]

$$(\cot \theta + \operatorname{cosec} \theta)^2 = \frac{1 + \cos \theta}{1 - \cos \theta}$$

End of Section A

START A NEW ANSWER BOOKLET

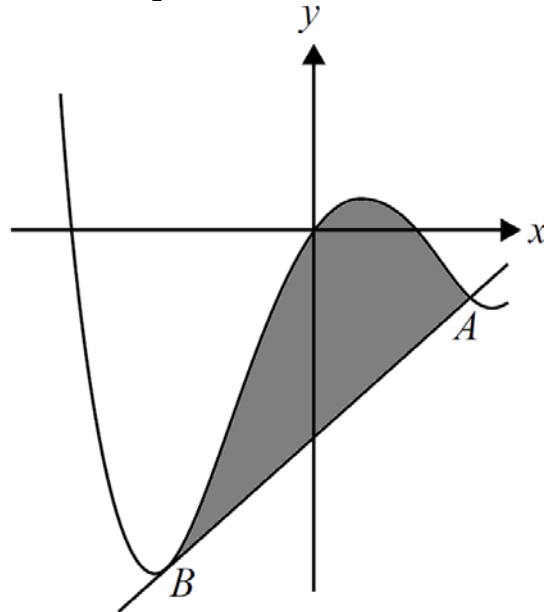
SECTION B [20 marks]

Marks
[2]

1. Solve $\cos \theta = \frac{1}{\sqrt{2}}$, for $0 \leq \theta \leq 2\pi$.

2. The diagram below shows the curve with equation

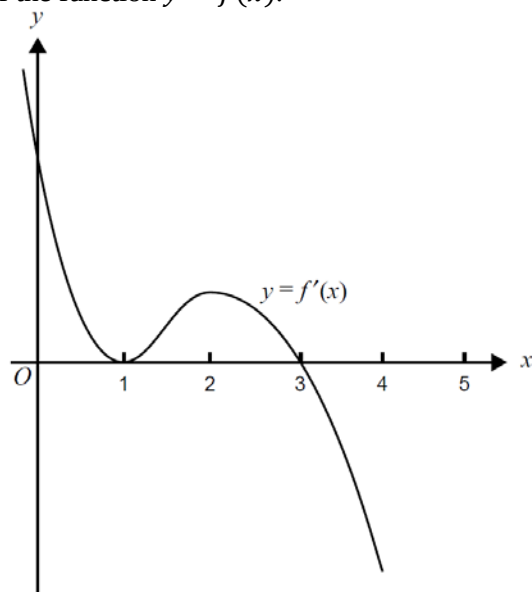
$$y = \frac{1}{2}(2x^4 - x^3 - 5x^2 + 3x).$$



The normal to the curve at A where $x = 1$ is a tangent to the curve at B.

- (i) Find the normal to the curve at A. [1]
- (ii) Use the factor theorem, or otherwise, find the coordinates of B. [3]
- (iii) Hence, find the area of the shaded region. [2]

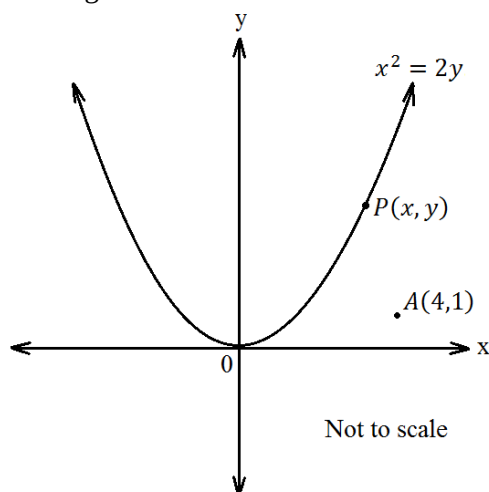
3. The diagram below shows the derivative of a function $y = f(x)$. Sketch a possible graph of the function $y = f(x)$. [2]



4. (i) Prove that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. [2]

- (ii) Hence, or otherwise, solve $\cos 3\theta + \sin 2\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$. [3]

5. The diagram below shows the graph of the parabola $x^2 = 2y$. The point $A(4,1)$ is outside the parabola while the point $P(x,y)$ is on the parabola as shown in the below diagram.



- (i) If D is the distance between the two points A and P , show that [1]

$$D^2 = \left(\frac{1}{2}x^2 - 1\right)^2 + (x - 4)^2$$

- (ii) Find the value of x that produces the minimum value for D in the equation in part (i). [3]

- (iii) Determine the minimum distance between A and P in exact form. [1]

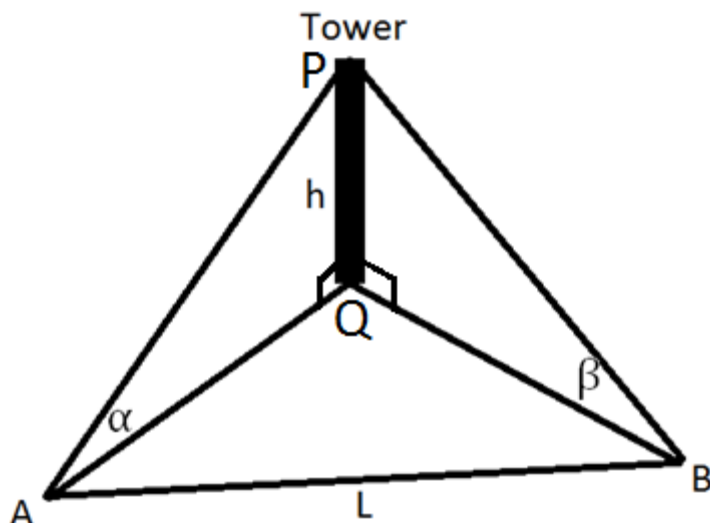
End of Section B

START A NEW ANSWER BOOKLET

SECTION C [16 marks]

Marks
[2]

1. If $x = c$ is a double zero of $P(x)$ i.e. $P(x) = (x - c)^2 Q(x)$, show that $x = c$ is a zero of $P'(x)$.
2. A tower has a height of h . At point A , the angle of elevation of the top of the tower is α , and at point B , the angle of elevation to the top of the tower is β . AB is separated by a distance of L and $\angle AQB = 60^\circ$.



[3]

[2]

- (i) Write an expression for h in terms of α, β and L .
 - (ii) If $\tan \alpha \tan \beta = x$ and $(\tan \alpha + \tan \beta)^2 = 3x + x^2$, show that $h = L$.
3.
 - (i) Express $\sqrt{3} \cos \theta - \sin \theta$ in the form $R \cos(\theta - \alpha)$, where $R > 0$ and α is an angle in radians. [2]
 - (ii) Hence, solve $\sqrt{3} \cos \theta - \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$. [2]
 4. $P(2p, p^2)$ and $Q(2q, q^2)$ are points on the parabola $x^2 = 4y$. The chord PQ always passes through the point $R(4, 0)$ when produced.
 - (i) Given the equation of chord PQ is $y = \frac{1}{2}(p + q)x - pq$, show that $pq = 2(p + q)$. [1]
 - (ii) M is the midpoint of chord PQ . Find the coordinates of M . [1]
 - (iii) Describe the locus of M , indicating any restrictions on x value. [3]

End of Section C

START A NEW ANSWER BOOKLET

SECTION D [14 marks]

Marks

1. Solve $\sin 2\theta - 3 \sin^2 \theta + 2 \sin \theta = 0$ for $-\pi \leq \theta \leq \pi$.

[3]

2. Solve for x ,

[3]

$$\frac{x}{1-x^2} \leq 0$$

3. Let α and β be the solutions of

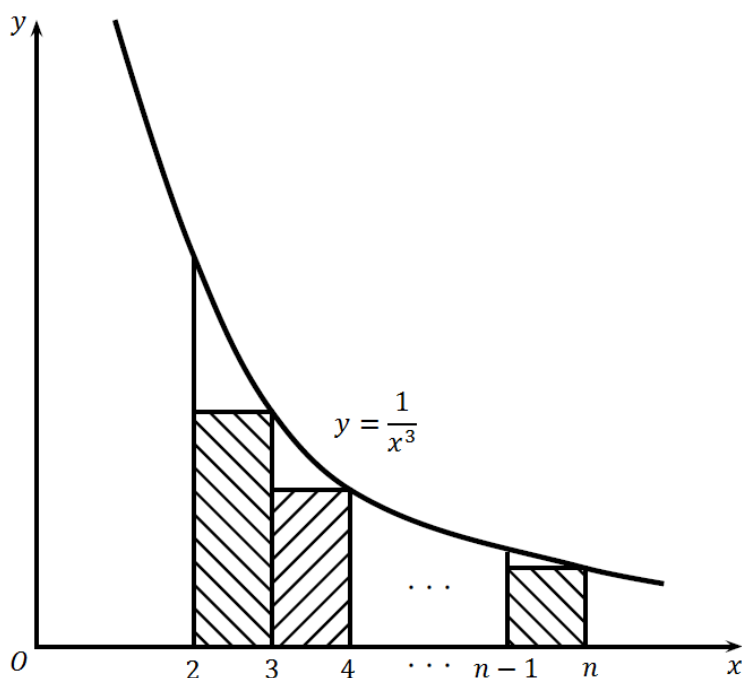
[4]

$$x^2 - (a+d)x + ad - bc = 0$$

Prove that α^3 and β^3 are the solutions of

$$x^2 - (a^3 + d^3 + 3abc + 3bcd)x + (ad - bc)^3 = 0$$

4. The diagram shows the curve $y = \frac{1}{x^3}$, for $x > 0$.



(i) Find an expression for the sum of areas of the shaded rectangles between $x = 2$ and $x = n$, where n is a positive integer.

[1]

(ii) Hence, or otherwise, prove that for any positive integer n ,

[3]

$$1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots + \frac{1}{n^3} < \frac{5}{4}$$

End of Section D
End of Exam

SECTION A

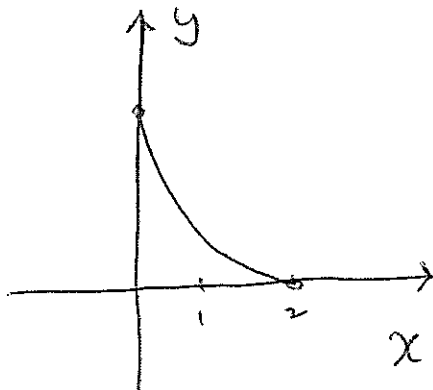
1. B
2. B
3. A
4. A
5. A
6. B
7. D
8. C
9. B
10. ~~C~~ B

11. $\tan \theta = \sqrt{3}$

$$\theta = \frac{\pi}{3} \pm n\pi$$

$(n \in \mathbb{Z}^+)$ [1]

12.



[2]

$$\begin{aligned} 13. \text{ LHS} &= (\cot \theta + \operatorname{cosec} \theta)^2 \\ &= \left(\frac{\cos \theta}{\sin \theta} + \frac{1}{\sin \theta} \right)^2 \\ &= \left(\frac{\cos \theta + 1}{\sin \theta} \right)^2 \\ &= \frac{(\cos \theta + 1)^2}{\sin^2 \theta} \\ &= \frac{(\cos \theta + 1)^2}{1 - \sin^2 \theta} \\ &= \frac{1 + \cos \theta}{1 - \cos \theta} \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} \text{OR } \text{RHS} &= \frac{1 + \cos \theta}{1 - \cos \theta} \times \frac{1 + \cos \theta}{1 + \cos \theta} \\ &= \frac{(1 + \cos \theta)^2}{1 - \cos^2 \theta} \\ &= \frac{1 + 2\cos \theta + \cos^2 \theta}{\sin^2 \theta} \\ &= \operatorname{cosec}^2 \theta + 2\operatorname{cosec} \theta \cot \theta + \cot^2 \theta \\ &= (\operatorname{cosec} \theta + \cot \theta)^2 \\ &= \text{LHS} \end{aligned}$$

[2]

Section B - Solutions

$$1. \cos \theta = \frac{1}{\sqrt{2}} \quad 0 \leq \theta \leq 2\pi$$

$$\theta = \frac{\pi}{4} \text{ or } \frac{7\pi}{4} \quad (2)$$

$$2(i) \quad y = \frac{1}{2}(2x^4 - x^3 - 5x^2 + 3x)$$

$$y' = \frac{1}{2}(8x^3 - 3x^2 - 10x + 3)$$

$$\text{At } x=1, y = -\frac{1}{2} \Rightarrow A = (1, -\frac{1}{2})$$

$$\text{At } x=1, y' = -1 = \text{gradient of tangent.}$$

$$\therefore \text{gradient of normal} = 1.$$

$$\text{Eqn of normal at A: } y + \frac{1}{2} = 1(x - 1) \\ y = x - \frac{3}{2} \quad (1)$$

$$(ii) \quad y = \frac{1}{2}(2x^4 - x^3 - 5x^2 + 3x) \quad (1) \\ y = x - \frac{3}{2} \quad (2)$$

$$\text{Sub in in (2)} \Rightarrow x - \frac{3}{2} = \frac{1}{2}(2x^4 - x^3 - 5x^2 + 3x) \\ 2x^4 - x^3 - 5x^2 + x + 3 = 0 \quad \checkmark$$

$$\text{Then } (x-1)(x+1) \text{ is a factor} \Rightarrow x^2 - 1 \text{ is a factor}$$

$$\begin{array}{r} x^2 - 1 \overline{) 2x^4 - x^3 - 5x^2 + x + 3} \\ \underline{2x^4 - 0x^3 - 2x^2} \\ -x^3 - 3x^2 + x \\ \underline{-x^3 + 0x^2 + 0x} \\ -3x^2 + x \\ \underline{-3x^2 + 0x + 3} \\ 0 \end{array} \quad \checkmark$$

$$\Rightarrow P(x) = (x-1)(x+1)(2x^2 - x - 3) = 0 \\ = (x-1)(x+1)^2(2x-3) = 0$$

2(ii) cont. Only one soln to left of A $\Rightarrow x = -1$

$$\therefore B = \left(-1, -\frac{5}{2}\right) \checkmark$$

(3)

$$(iii) \text{ Area} = \int_{-1}^1 \frac{1}{2} (2x^4 - x^3 - 5x^2 + 3x) - \left(x - \frac{3}{2}\right) dx$$

$$= \int_{-1}^1 \left(x^4 - \frac{1}{2}x^3 - \frac{5}{2}x^2 + \frac{1}{2}x + \frac{3}{2}\right) dx \checkmark$$

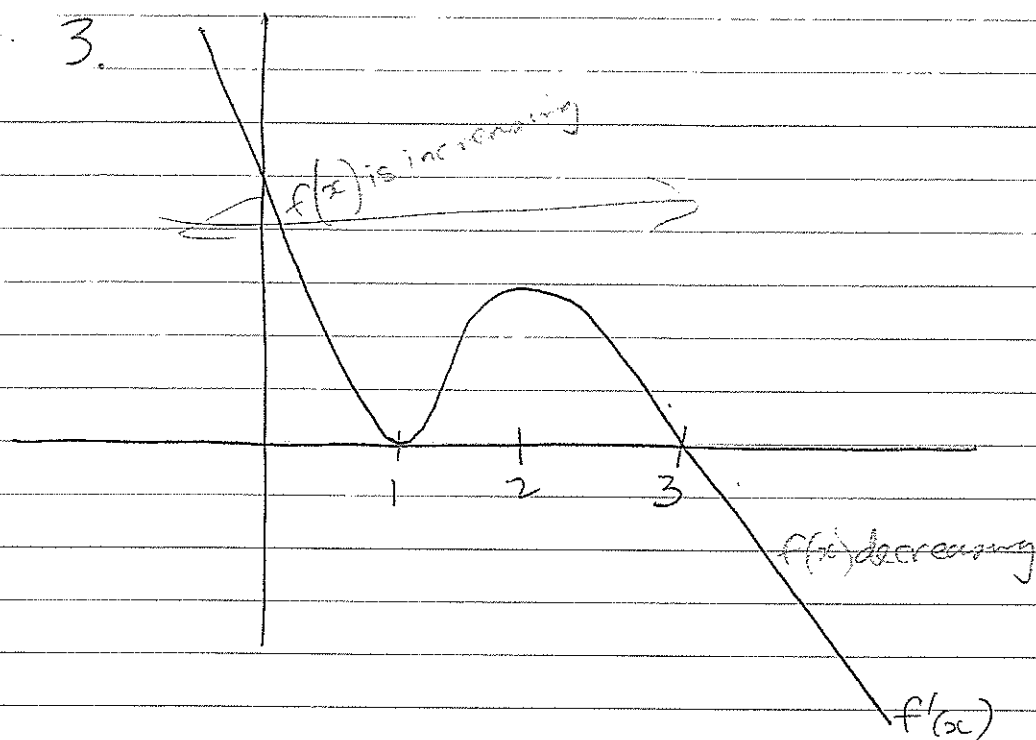
$$= \left[\frac{x^5}{5} - \frac{x^4}{8} - \frac{5x^3}{6} + \frac{x^2}{4} + \frac{3x}{1} \right]_{-1}^1$$

$$= \left(\frac{1}{5} - \frac{1}{8} - \frac{5}{6} + \frac{1}{4} + \frac{3}{2} \right) - \left(-\frac{1}{5} - \frac{1}{8} + \frac{5}{6} + \frac{1}{4} - \frac{3}{2} \right)$$

$$= \frac{26}{15} \text{ or } 1.7\bar{3} \text{ units}^2 \checkmark$$

(2)

3.



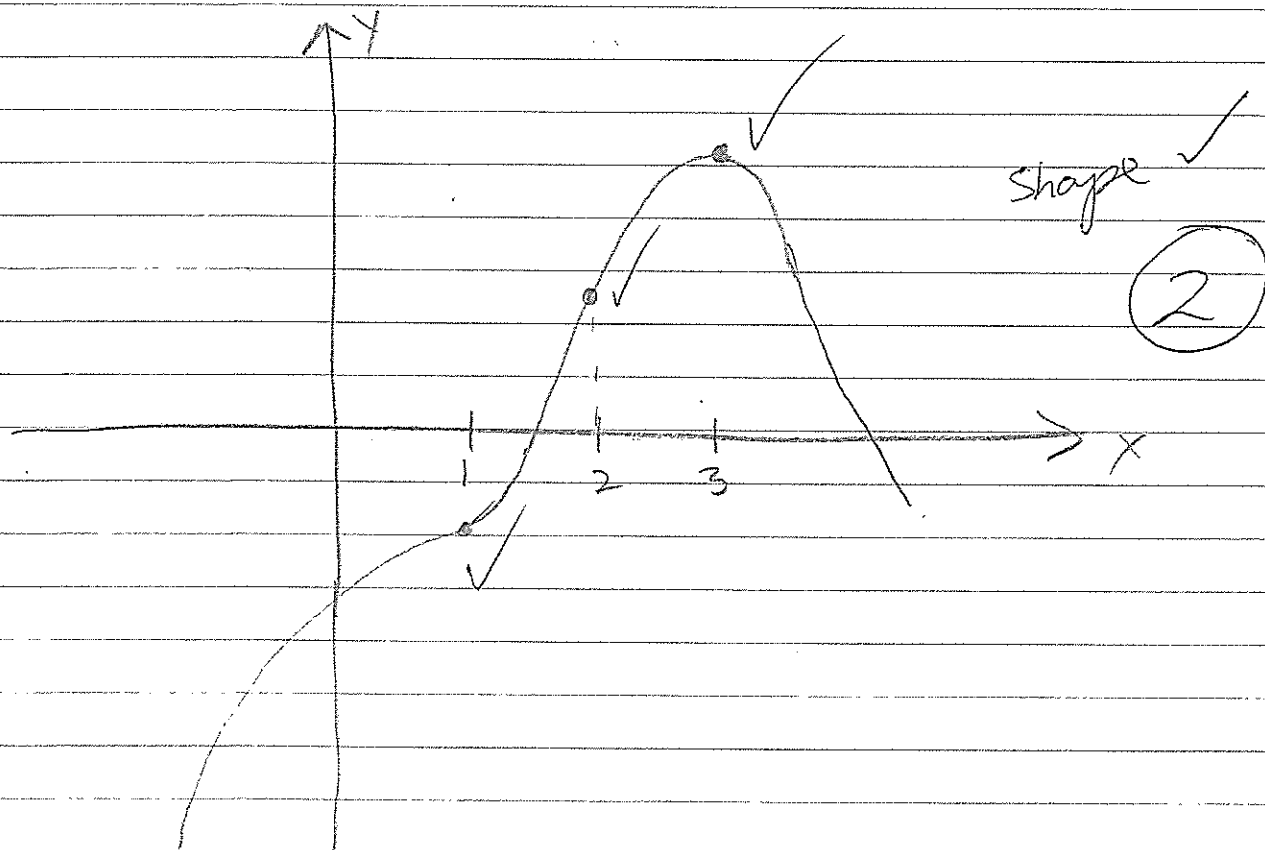
$f'(x) = 0 \Rightarrow$ turn pts at $x=1, x=3 \leftarrow$ PHI $\leftarrow$ +ve, 0, -ve
max + p.

$f''(x) = 0 \Rightarrow$ change in concavity.

$\therefore$ change in concavity at $x=2$

When $f'(x) > 0$, $f(x)$ increasing

$f'(x) < 0$, $f(x)$ decreasing



4(i) $\frac{\text{Prove}}{\cos} 3\theta = 4\cos^3\theta - 3\cos\theta$

$$\begin{aligned}\cos(2\theta + \theta) &= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta \\ &= (2\cos^2\theta - 1)\cos\theta - 2\sin\theta \cos\theta \sin\theta \\ &= 2\cos^3\theta - \cos\theta - 2\cos\theta(1 - \cos^2\theta) \\ &= 2\cos^3\theta - \cos\theta - 2\cos\theta + 2\cos^3\theta \\ &= 4\cos^3\theta - 3\cos\theta\end{aligned}$$

(2)

(ii) Solve $\cos 3\theta + \sin 2\theta = \cos \theta$ for $0 \leq \theta \leq 2\pi$

Now $4\cos^3\theta + 2\sin\theta \cos\theta - 4\cos\theta = 0$

from (i)

$$\cos\theta(4\cos^2\theta + 2\sin\theta - 4) = 0$$

$$2\cos\theta(2\cos^2\theta + \sin\theta - 2) = 0$$

$$2\cos\theta(2(1 - \sin^2\theta) + \sin\theta - 2) = 0$$

$$2\cos\theta(2 - 2\sin^2\theta + \sin\theta - 2) = 0$$

$$2\cos\theta(\sin\theta(1 - 2\sin\theta)) = 0$$

$$\Rightarrow \cos\theta = 0 \text{ or } \sin\theta = 0 \text{ or } \sin\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2} \text{ or } \theta = 0, \pi, 2\pi \text{ or } \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

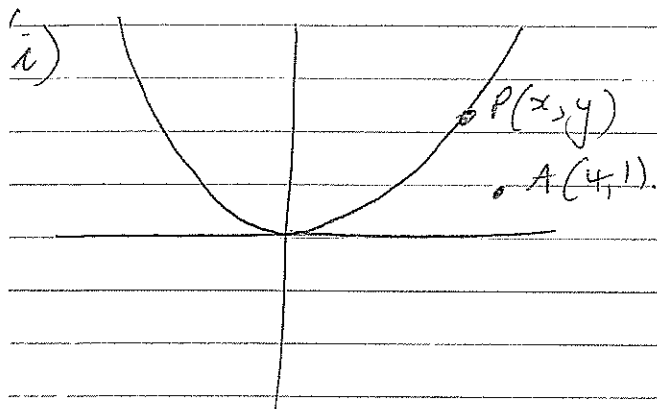
$$\Rightarrow \theta = \frac{\pi}{2}, \frac{3\pi}{2}, 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6}$$

(3)

$$5. \quad x^2 = 2y.$$

$$\begin{aligned} 4a &= 2 \\ a &= \frac{1}{2} \end{aligned}$$

$A(4, 1)$ ext. pt.



$$AP^2 = D^2 = (x-4)^2 + (y-1)^2$$

$$\text{but } y = \frac{x^2}{2}$$

$$\Rightarrow D^2 = (x-4)^2 + \left(\frac{x^2}{2} - 1\right)^2 \quad \checkmark \textcircled{1}$$

$$D^2 = (x-4)^2 + \left(\frac{1}{2}x^2 - 1\right)^2 \quad \#$$

$$(ii) \quad D^2 = (x-4)^2 + \left(\frac{1}{2}x^2 - 1\right)^2$$

$$\begin{aligned} (D^2)' &= 2(x-4) + 2\left(\frac{1}{2}x^2 - 1\right)x. \quad \checkmark \\ &= 2x - 8 + x^3 - 2x \end{aligned}$$

$$(D^2)' = x^3 - 8$$

$$\text{For t.p. } (D^2)' = 0 \Rightarrow x^3 = 8 \quad \checkmark$$

$$\underline{x = \pm 2.}$$

$\textcircled{3}$

$$(D^2)'' = 3x^2$$

$$\text{At } x=2, (D^2)'' = 12 > 0 \Rightarrow \text{min at } x=2.$$

$$x=-2, (D^2)'' = -12 < 0 \Rightarrow \text{max at } x=-2. \quad \checkmark$$

$\therefore$ Value of $x = 2$ to give min. for D .

$$\begin{aligned} (iii) \quad D^2 &= (2-4)^2 + (2-1)^2 \\ D^2 &= 4 + 1 \end{aligned}$$

$$\rightarrow D = \sqrt{5} \quad \checkmark \textcircled{1}$$

Section C

(1)

$$P(x) = (x - c)^2 Q(x)$$

$$P'(x) = 2(x - c)Q(x) + (x - c)^2 Q'(x)$$

$$P'(x) = (x - c)[2Q(x) + (x - c)Q'(x)]$$

$$P'(c) = (c - c)[2Q(c) + (c - c)Q'(c)]$$

$$P'(c) = 0$$

(2)(i)

$$\tan \alpha = \frac{h}{AQ} \Rightarrow AQ = \frac{h}{\tan \alpha}$$

$$\tan \beta = \frac{h}{BQ} \Rightarrow BQ = \frac{h}{\tan \beta}$$

$$L^2 = AQ^2 + BQ^2 - 2AQ \times BQ \cos 60^\circ$$

$$L^2 = \frac{h^2}{\tan^2 \alpha} + \frac{h^2}{\tan^2 \beta} - \frac{h^2}{\tan \alpha \tan \beta}$$

$$L^2 = h^2 \left[\frac{\tan^2 \alpha + \tan^2 \beta}{\tan^2 \alpha \tan^2 \beta} - \frac{\tan \alpha \tan \beta}{\tan^2 \alpha \tan^2 \beta} \right]$$

$$L^2 = h^2 \left[\frac{(\tan \alpha + \tan \beta)^2 - 3 \tan \alpha \tan \beta}{\tan^2 \alpha \tan^2 \beta} \right]$$

$$h = \frac{L \tan \alpha \tan \beta}{\sqrt{(\tan \alpha + \tan \beta)^2 - 3 \tan \alpha \tan \beta}}$$

(ii)

$$h = \frac{xL}{\sqrt{3x + x^2 - 3x}}$$

$$h = L$$

(3)(i)

$$R \cos(\theta - \alpha) = R \cos \alpha \cos \theta + R \sin \alpha \sin \theta$$

$$R \cos \alpha = \sqrt{3}$$

$$R \sin \alpha = -1$$

$$R^2 = 4$$

$$R = 2$$

$$\tan \alpha = -\frac{1}{\sqrt{3}}$$

$$\alpha = \frac{11\pi}{6} \text{ or } -\frac{\pi}{6} \text{ 4th quadrant}$$

$$2 \cos \left(\theta - \frac{11\pi}{6} \right) \text{ or}$$

$$2 \cos \left(\theta + \frac{\pi}{6} \right)$$

(ii)

$$2 \cos \left(\theta - \frac{11\pi}{6} \right) = 1$$

$$\theta - \frac{11\pi}{6} = \cos^{-1} \left(\frac{1}{2} \right) \text{ where } -\frac{11\pi}{6} \leq \theta - \frac{11\pi}{6} \leq \frac{\pi}{6}$$

$$\theta - \frac{11\pi}{6} = -\frac{\pi}{3}, -\frac{5\pi}{3}$$

$$\theta = \frac{3\pi}{2}, \frac{\pi}{6}$$

(4)(i)

$$y = \frac{1}{2}(p + q)x - pq$$

$$0 = 2(p + q) - pq$$

$$pq = 2(p + q)$$

(ii)

$$M \left(p + q, \frac{p^2 + q^2}{2} \right)$$

(iii)

$$x = p + q$$

$$y = \frac{(p + q)^2 - 2pq}{2}$$

$$y = \frac{x^2 - 4x}{2}$$

$$y = \frac{x^2}{2} - 2x$$

$$\text{Restrictions where } 0 \leq 4p - p^2$$

$$\text{that is } x \leq 0, x \geq 8$$

SECTION D

Q1 $\sin 2\theta - 3\sin^2 \theta + 2\sin \theta = 0$, $|\theta| \leq \pi$

$$\therefore 2\sin \theta \cos \theta - 3\sin^2 \theta + 2\sin \theta = 0$$

$$\sin \theta (2\cos \theta - 3\sin \theta + 2) = 0$$

$$\therefore \sin \theta = 0 \quad \text{OR} \quad 2\cos \theta - 3\sin \theta + 2 = 0.$$

$$\theta = 0, \pm\pi \quad \text{OR} \quad \frac{2(1-t^2)}{1+t^2} - \frac{6t}{1+t^2} + 2 = 0$$

$$2 - 2t^2 - 6t + 2 + 2t^2 = 0$$

$$6t = 4$$

$$t = \frac{2}{3}$$

$$\therefore \tan \frac{\theta}{2} = \frac{2}{3}$$

$$\frac{\theta}{2} = \tan^{-1} \frac{2}{3}$$

$$\theta = 2\tan^{-1} \frac{2}{3}$$

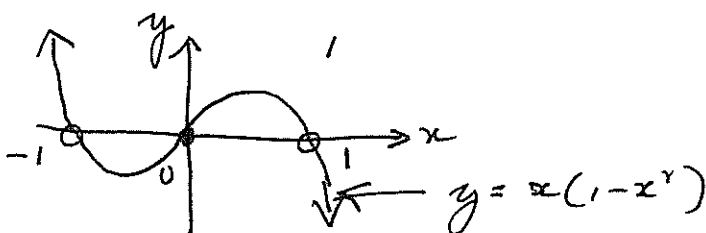
$$\therefore \boxed{\theta = 0, \pm\pi, 2\tan^{-1} \frac{2}{3}}$$

$$\sim 1.176.$$

Q2. $\frac{x}{1-x^2} \leq 0$

$$\therefore x(1-x^2) \leq 0$$

(MULTIPLY BOTH
SIDES BY $(1-x^2)^2$)



from graph.

$$\boxed{-1 < x \leq 0, x > 1.}$$

Q3.

$$\text{now } \alpha + \beta = a + d \quad \text{--- (A)}$$

$$\alpha\beta = ad - bc \quad \text{--- (B)}$$

$$\text{now } \alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)$$

$$= (\alpha + \beta)((\alpha + \beta)^2 - 3\alpha\beta)$$

$$\text{--- (A)}$$

$$= (a + d)((a + d)^2 - 3(ad - bc))$$

$$= (a + d)(a^2 + 2ad + d^2 - 3ad + 3bc)$$

$$\text{--- (A)}$$

$$= a^3 + 2a^2d + ad^2 - 3a^2d + 3abc$$

$$+ da^2 + 2ad^2 + d^3 - 3ad^2 + 3bcd$$

$$= a^3 + d^3 + 3abc + 3bcd$$

now required quadratic is $x^2 - (\alpha^3 + \beta^3)x + \alpha^3\beta^3$.

$$\text{ie. } \boxed{x^2 - (a^3 + d^3 + 3abc + 3bcd)x + (ad - bc)^3 = 0}$$

$$(\text{NB. from (B) } (\alpha\beta)^3 = \alpha^3\beta^3 = (ad - bc)^3)$$

Q4.

$$(i) \quad 1 \times \frac{1}{3^3} + 1 \times \frac{1}{4^3} + 1 \times \frac{1}{5^3} + \dots + 1 \times \frac{1}{n^3} = 0$$

$$\left[\text{or } \sum_{r=3}^n \frac{1}{r^3} \right]$$

$$\begin{aligned} (ii) \quad \text{now } \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \dots + \frac{1}{n^3} &< \int_2^n x^{-3} dx \\ &= \left[\frac{-1}{2x^2} \right]_2^n \\ &= \frac{-1}{2n^2} + \frac{1}{2^3} \\ &= \frac{1}{8} - \frac{1}{2n^2} \end{aligned}$$

$$\begin{aligned} \therefore 1 + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \dots + \frac{1}{n^3} &< \frac{1}{8} - \frac{1}{2n^2} + 1 + \frac{1}{2^3} \\ &= \frac{5}{4} - \frac{1}{2n^2} \end{aligned}$$

$$\therefore \left[1 + \frac{1}{2^3} + \frac{1}{3^3} + \dots + \frac{1}{n^3} < \frac{5}{4} - \frac{1}{2n^2} \right. \\ \left. < \frac{5}{4} \quad \left(\frac{1}{2n^2} > 0 \right) \right]$$